

भारतीय सूचना प्रौद्योगिकी संस्थान गुवाहाटी

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY GUWAHATI

CS 360: Machine Learning Lab
Practice Assignment 9

Instructions: This is only for practice. Complete it by 12:00 PM today. Your completion will be reviewed by the Teaching Assistants.

1. Write a program to perform classification on OR-gate and NAND-gate using Perceptron learning algorithm (without using in-built python packages): Consider the following for implementation of the above,

   - Activation function: Step function
   - w= random initialization.
   - Number of epochs:2
   - train:test split: 80:20

   (a) Report the class-wise accuracy, precision, recall for training and testing samples, after 2 epochs.

2. An e-commerce company wants to predict whether a customer will purchase a product based on browsing behaviour and customer characteristics, based on the following attributes with the given values:

   - Age group: Young, middle, senior
   - Income: High, medium, low
   - : Browsing time:long, short
   - previous purchase: Yes,no
   - Purchase (Y/ target): Yes,no

   (a) Implement a Naive Bayes classifier (without using in-built python packages) to predict the target(Purchase) for the test sample: (Age group=middle, Income= High, browsing time=long, previous purchase= Yes).

   (b) Evaluate the deigned classifier using a confusion matrix and report the accuracy, class-wise accuracy, precision and recall.

3. Create a synthetic dataset for a healthcare organization that wants to classify individuals into diabetic and non-diabetic categories based on the following continuous-valued features as below:

   - Glucose level
   - BMI
   - Age

(a) Implement Gaussian Naive Bayes classifier, (without using in-built python packages) to predict the target(diabetic:yes,no) for the test sample: (Glucose level-142, BMI= 29, Age=42).

(b) Evaluate the deigned classifier using a confusion matrix and report the accuracy, class-wise accuracy, precision and recall

(c) Compare the predictions and class conditional probabilities obtained from both implementations (2(a), 3(a)) and report your findings/ observations.